

# SETS — Short Revision Notes

Understand Level: Explain Ideas & Concepts

## 1. Understanding What a Set Is

| Concept              | Explanation                                 | Example                                             |
|----------------------|---------------------------------------------|-----------------------------------------------------|
| Definition           | A collection of well-defined objects        | {1, 2, 3, 4, 5}                                     |
| Well-Defined         | You can clearly decide if an object belongs | "Good students" is NOT a set; "Grade 9 students" IS |
| Elements             | The objects that belong to a set            | In {2,4,6,8}, the elements are 2,4,6,8              |
| Order Doesn't Matter | Sets are unordered collections              | {1,2,3} = {3,2,1}                                   |

### Understanding "Well-Defined"

| Collection                      | Is it a Set? | Why / Why Not                        |
|---------------------------------|--------------|--------------------------------------|
| Multiples of 4 between 0 and 30 | YES ✓        | Clear: {4,8,12,16,20,24,28}          |
| Districts of Sri Lanka          | YES ✓        | Clear: 25 districts                  |
| Students good at mathematics    | NO ✗         | "Good" is subjective                 |
| Triangular numbers              | YES ✓        | Clear: {1,3,6,10,15,...}             |
| The 10 largest integers         | NO ✗         | No such thing as 10 largest integers |

Key: a set must have a CLEAR rule so everyone agrees on what belongs and what doesn't.

## 2. Understanding How to Write Sets

| Method           | Explanation                      | Example                                |
|------------------|----------------------------------|----------------------------------------|
| Descriptive      | Describes the set in words       | {Multiples of 5 between 0 and 20}      |
| Roster (Listing) | Lists elements in curly brackets | {5,10,15,20}                           |
| Set-Builder      | Uses a rule to describe elements | {x   x is a multiple of 5, 0 < x < 20} |
| Venn Diagram     | Shows elements in a circle       | Circle with 5,10,15,20 inside          |

Set: Even numbers between 0 and 10

Descriptive: {Even numbers between 0 and 10}

Roster: {2, 4, 6, 8}

Set-Builder: {x | x is even, 0 < x < 10}

Venn Diagram: Circle containing 2,4,6,8

### 3. Understanding Finite and Infinite Sets

| Type     | Explanation                           | Example                                    |
|----------|---------------------------------------|--------------------------------------------|
| Finite   | Elements can be counted completely    | $\{2,4,6,8,10\} \rightarrow 5$ elements    |
| Infinite | Elements cannot be counted completely | $\{2,4,6,8,10,\dots\} \rightarrow$ endless |

| Set                                 | Type     | Why?                          |
|-------------------------------------|----------|-------------------------------|
| {Mon, Tue, Wed, Thu, Fri, Sat, Sun} | Finite   | 7 days                        |
| {Multiples of 3}                    | Infinite | Goes on forever: 3,6,9,12,... |
| {Letters in "MATHEMATICS"}          | Finite   | 11 letters (some repeated)    |
| {Prime numbers}                     | Infinite | 2,3,5,7,11,13,...             |
| {Students in your class}            | Finite   | Countable number of students  |

Key: if you can finish counting all elements  $\rightarrow$  FINITE. If the counting never ends  $\rightarrow$  INFINITE.

### 4. Understanding Equal and Equivalent Sets

| Concept         | Explanation               | Example                                            |
|-----------------|---------------------------|----------------------------------------------------|
| Equal Sets      | Exactly the SAME elements | $A=\{1,2,3\}$ , $B=\{3,2,1\} \rightarrow A=B$      |
| Equivalent Sets | Same NUMBER of elements   | $A=\{1,2,3\}$ , $B=\{4,5,6\} \rightarrow A \sim B$ |

| Set A       | Set B         | Equal? | Equivalent? | Why?                            |
|-------------|---------------|--------|-------------|---------------------------------|
| $\{1,2,3\}$ | $\{1,2,3\}$   | YES ✓  | YES ✓       | Same elements & number          |
| $\{1,2,3\}$ | $\{3,2,1\}$   | YES ✓  | YES ✓       | Order doesn't matter            |
| $\{1,2,3\}$ | $\{4,5,6\}$   | NO ✗   | YES ✓       | Different elements, same number |
| $\{1,2,3\}$ | $\{1,2,3,4\}$ | NO ✗   | NO ✗        | Different elements & number     |

Key relationships: all equal sets are equivalent, but not all equivalent sets are equal.

## 5. Understanding Subsets

| Concept          | Explanation                            | Example                       |
|------------------|----------------------------------------|-------------------------------|
| Subset           | All elements of one set are in another | $\{2,4\} \subset \{2,4,6,8\}$ |
| Proper Subset    | A subset not equal to the original     | $\{2,4\} \subset \{2,4,6,8\}$ |
| Empty Set Subset | $\phi$ is a subset of EVERY set        | $\phi \subset A$ always       |

| Set A       | Set B         | $A \subset B?$ | Why?                            |
|-------------|---------------|----------------|---------------------------------|
| $\{2,4\}$   | $\{2,4,6,8\}$ | YES ✓          | All elements of A are in B      |
| $\{2,4\}$   | $\{1,2,3,4\}$ | YES ✓          | 2 and 4 are in B                |
| $\{2,7\}$   | $\{1,2,3,4\}$ | NO ✗           | 7 is not in B                   |
| $\{1,2,3\}$ | $\{1,2,3\}$   | YES ✓          | Every set is a subset of itself |

Set  $X = \{1,2\}$  → All subsets:  $\phi, \{1\}, \{2\}, \{1,2\}$

Number of subsets =  $2^n$  ( $n$  = number of elements)

For  $\{1,2\}$ :  $2^2 = 4$  subsets

For  $\{3,5,7\}$ :  $2^3 = 8$  subsets

## 6. Understanding the Universal Set

| Concept       | Explanation                               | Example                         |
|---------------|-------------------------------------------|---------------------------------|
| Universal Set | Contains ALL elements under consideration | $\epsilon = \{1,2,3,\dots,10\}$ |
| Subsets       | All sets are subsets of the universal set | All sets are inside $\epsilon$  |

Study: Students in a school

Universal Set ( $\epsilon$ ) = {All students in the school}

Subset A = {Grade 9 students}

Subset B = {Female students}

Subset C = {Students who play cricket}

All subsets sit inside the universal set.

| Sets Being Considered                   | Suitable Universal Set        |
|-----------------------------------------|-------------------------------|
| {Grade 9 students}, {Grade 10 students} | {All students in the school}  |
| {Even numbers}, {Odd numbers}           | {All integers}                |
| {Red cards}, {Black cards}              | {All cards in a deck}         |
| {Vowels}, {Consonants}                  | {All letters in the alphabet} |

## 7. Understanding Intersection ( $\cap$ )

| Concept      | Explanation                                 |
|--------------|---------------------------------------------|
| Intersection | Elements COMMON to both sets ( $A \cap B$ ) |
| Symbol       | $\cap$ — looks like an upside-down U        |
| Meaning      | "AND" — must be in both sets                |

$A = \{1, 2, 3, 4, 5\}$        $A = \{2, 4, 6, 8\}$   
 $B = \{4, 5, 6, 7, 8\}$        $B = \{1, 3, 5, 7\}$   
 $A \cap B = \{4, 5\}$        $A \cap B = \emptyset$  (disjoint)  
(4 and 5 are in BOTH sets)

Venn diagram:  $A \cap B$  is the overlapping region of the two circles.

## 8. Understanding Union ( $\cup$ )

| Concept | Explanation                                        |
|---------|----------------------------------------------------|
| Union   | All elements in EITHER or BOTH sets ( $A \cup B$ ) |
| Symbol  | $\cup$ — looks like a cup                          |
| Meaning | "OR" — in A or B or both                           |

$A = \{1, 2, 3, 4, 5\}$        $A = \{2, 4, 6\}$   
 $B = \{4, 5, 6, 7, 8\}$        $B = \{1, 3, 5\}$   
 $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$        $A \cup B = \{1, 2, 3, 4, 5, 6\}$   
(all elements from BOTH sets, no duplicates)

Venn diagram:  $A \cup B$  is every shaded part of both circles.

## 9. Understanding Complement ( $A'$ )

| Concept    | Explanation                       |
|------------|-----------------------------------|
| Complement | Elements in $e$ NOT in A ( $A'$ ) |
| Meaning    | "Everything except A"             |
| Reading    | "A prime" or "A complement"       |

$e = \{1, 2, 3, 4, 5, 6, 7\}$   
 $A = \{2, 4, 6\}$   
 $A' = \{1, 3, 5, 7\}$  (elements in  $e$  that are NOT in A)

Important:  $A \cup A' = e$        $A \cap A' = \emptyset$

Venn diagram:  $A'$  is everything inside the rectangle but outside circle A.

## 10. Understanding Disjoint Sets

| Concept  | Explanation                                        |
|----------|----------------------------------------------------|
| Disjoint | Sets with NO common elements ( $A \cap B = \phi$ ) |
| Meaning  | They have nothing in common                        |

$$A = \{1, 3, 5, 7\}$$

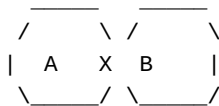
$$B = \{2, 4, 6, 8\}$$

$$A \cap B = \phi \quad (\text{no element is in both sets})$$

Venn diagram: two separate circles that do not overlap.

## 11. Understanding Venn Diagrams

| Situation        | Venn Diagram                 | Meaning                               |
|------------------|------------------------------|---------------------------------------|
| A and B overlap  | Two circles overlapping      | $A \cap B \neq \phi$ (share elements) |
| A and B disjoint | Two separate circles         | $A \cap B = \phi$ (no common)         |
| B is subset of A | Circle B inside circle A     | $B \subset A$                         |
| $A = B$          | Two identical circles        | Same elements                         |
| Complement       | Rectangle with circle inside | $A'$ outside the circle               |



$X = A \cap B$  (elements in both)  
 Only A / Only B on each side  
 Outside both circles =  $(A \cup B)'$

## 12. Common Misunderstandings to Avoid

| Misunderstanding                          | Why It's Wrong               | Correct Understanding                           |
|-------------------------------------------|------------------------------|-------------------------------------------------|
| $\{2,4,6\}$ and $\{6,4,2\}$ are different | Order doesn't matter         | They are EQUAL sets                             |
| Equal = equivalent                        | Same elements vs same number | Equal = same elements; Equivalent = same number |
| $\phi$ has one element                    | $\phi$ has NO elements       | $n(\phi) = 0$                                   |
| $\epsilon$ is always $\{1,2,3,\dots\}$    | Depends on context           | $\epsilon$ = all elements under consideration   |
| All subsets are proper subsets            | A set is a subset of itself  | Proper subset $\neq$ subset                     |

## 13. Check Your Understanding

| No. | Question                                      | Explanation                                     |
|-----|-----------------------------------------------|-------------------------------------------------|
| 1   | What is a set?                                | A well-defined collection of objects            |
| 2   | What makes a collection "well-defined"?       | A clear rule for membership                     |
| 3   | Difference between finite and infinite sets?  | Finite = countable, Infinite = uncountable      |
| 4   | Difference between equal and equivalent sets? | Equal = same elements, Equivalent = same number |
| 5   | What is a subset?                             | All elements of one set are in another          |
| 6   | What is a universal set?                      | Set of all elements under consideration         |
| 7   | What is intersection?                         | Elements common to both sets                    |
| 8   | What is union?                                | All elements in either or both sets             |
| 9   | What is complement?                           | Elements in universal set not in A              |
| 10  | What are disjoint sets?                       | Sets with no common elements                    |

## 14. Key Understandings to Remember

- Sets are well-defined collections.
- $\{1,2,3\} = \{3,2,1\}$  (order doesn't matter).
- FINITE = countable, INFINITE = uncountable.
- EQUAL sets -> same elements.  
EQUIVALENT sets -> same number.
- $\phi$  (null set) has no elements;  $n(\phi) = 0$ .
- $\epsilon$  (universal set) contains all elements.
- $A \subset B$  = all elements of A are in B.
- $A \cap B$  = elements in both.
- $A \cup B$  = elements in either or both.
- $A'$  = elements not in A.
- A and B disjoint = no common elements.

Revision Tip: Understand the meaning of each set operation and symbol. Visualize using Venn diagrams — they make the relationships much clearer.